Participatory Research Report on Water Crisis and Health Problems in a Marma Community of Bandarban

Introduction

This report shares the findings of a participatory research study done with the Marma Indigenous community in Amtoli Marma Para, Suwalok Union, Bandarban Sadar, Bangladesh. The study looked at how community members deal with water shortages and water-related diseases during summer when natural water sources dry up (bdnews24.com, 2026). Using three participatory methods – Farmers Field School (FFS), Transect Walk, and Resource Mapping – the research worked with community members as active partners in sharing their knowledge, analyzing problems, and finding solutions (Cornwall & Jewkes, 1995). The findings show that the community increasingly depends on shallow hand-dug pits called "Kua" beside streams for drinking water during dry seasons, even though this water is not safe (The Daily Post, 2024). The research found key coping methods including rainwater collection, use of traditional water sources, and community-level adaptation practices (Datta et al., 2025). An action plan made together with community members suggests short-term and long-term solutions, including deeper ring wells, community water filters, and planting more trees.

Research Objectives

To understand how the Marma community of Amtoli Para in Bandarban copes with water crises and water-related diseases during summer months when natural water sources dry up (bdnews24.com, 2026).

No.	Objective
1	To identify and document the current coping strategies used by community members to deal with water shortage during summer (bdnews24.com, 2026)
2	To understand the types of water-related diseases community members experience and their causes (The Daily Post, 2024)
3	To document indigenous and traditional knowledge related to water finding and water management in the Marma community (Datta et al., 2025)
4	To identify structural, environmental, and governance barriers that prevent access to safe water (bdnews24.com, 2026)
5	To co-develop an action plan with community members for short-term and long-term solutions to the water crisis (Minkler & Wallerstein, 2011)

Research Design

Why We Chose This Community

We chose Amtoli Marma Para in Suwalok Union of Bandarban Sadar for three main reasons. First, this community faces severe water shortage every year. People reported that from March to May, water sources become very low or completely dry (bdnews24.com, 2026). Second, the Marma Indigenous community has rich traditional knowledge about water management that has not been written down or shared widely (Datta et al., 2025). Third, the Department of Public Health Engineering has called this area an "unsuccessful zone" for normal tubewells. This makes it a good place to study alternative ways people cope with water shortage (bdnews24.com, 2026).

The community has about 160 families and a history of more than 300 years (Datta et al., 2025). Like many Marma communities in Bandarban, Amtoli Para faces serious threats from climate change including flash floods, droughts, and landslides (Datta et al., 2025).

How We Entered the Community and Built Trust

Building trust with the Marma community required a culturally respectful approach (Chambers, 2008). We took the following steps:

First Contact with Local Leaders
Before starting any research, we met with the Karbari Menpom Marma (traditional community leader) and Union Parishad representatives. Ukynu Marma, the Union Parishad Chairman, gave us guidance on local protocols and expectations.

Multiple Informal Visits
We have visited the community three times over two weeks without doing any formal research. These visits involved sitting at local tea stalls, attending community gatherings, and simply listening and travelling near the water sources like waterfalls.

Community Meeting to Explain the Purpose
We organized a small meeting with the presence with four to five people where we clearly explained the research purpose: to understand and document how the community copes with water shortages and to work together to find solutions. We assured community members that they would own the findings.

Ethics and Consent

This research followed the core ethical rules of participatory research (Israel et al., 2017). We obtained informed consent orally in the Marma language using a simple script. Key ethical practices included:

Principle	How We Did It
Voluntary Participation	No one was forced to participate; anyone could stop at any time (Israel et al., 2017)
Confidentiality	We did not link individual names to sensitive information
Do No Harm	Research did not interfere with people's daily water collection activities
Benefit Sharing	We will share findings back through community meetings and printed materials (Minkler & Wallerstein, 2011)
Cultural Respect	All activities were scheduled around community routines and religious practices

How We Did the Research (Methodology)

Farmers Field School (FFS)

The Farmers Field School method is a learning approach based on discovery and observation. It helps farmers analyze their local problems and develop their own solutions (Food and Agriculture Organization, 2017). Unlike top-down approaches where experts give advice, FFS treats farmers as experts who share knowledge with each other and with researchers (Isubikalu, 2007).

How We Did It:

Over four weekly sessions, 15 community members (8 women, 7 men) took part in FFS activities focused on water and health issues.

Session 1: Identifying Problems
Participants identified and ranked their most pressing water-related problems using a ranking exercise (Chambers, 2008).

Problem	Severity (1-5)	Frequency (1-5)	Health Impact (1-5)	Total
Water shortage in summer	5	5	4	14
Dirty water sources	4	4	5	13
Distance to water sources	4	5	3	12
Tube well failure	4	4	3	11

Session 2: Documenting Traditional Knowledge
Elder community members shared traditional water management practices (Datta et al., 2025):

Traditional pit digging (Kua) techniques passed down through generations

Signs for finding water during dry seasons (certain plants, soil moisture patterns)

Rainwater collection using locally available materials

Session 3: Understanding Health Impacts
Community members discussed waterborne diseases they commonly experience (The Daily Post, 2024):

Diarrhea and dysentery (reported by 70% of participating families)

Skin infections (reported by 55%)

Typhoid (reported by 30% over the past two years)

Session 4: Brainstorming Solutions
Participants identified possible solutions including deeper ring wells, water filters, and planting more trees (bdnews24.com, 2026).

Transect Walk

The transect walk involved walking through the community in a straight line to observe land use patterns, water sources, and environmental problems (Chambers, 2008).

Planning the Walk Route
We planned a 2.5 kilometer walk route with help from community elders. The route covered:

The Suwalok Stream area

Farming areas (jhum cultivation fields)

Residential areas

Forested hill slopes

Existing water structures (tube wells, ring wells, pits)

What We Observed:

What We Looked At	What We Found
Land Use	Mostly jhum farming on slopes; small gardens near houses (Datta et al., 2025)
Water Sources	One working tube well, two ring wells (one at Buddhist temple), Suwalok Stream (seasonal), many hand-dug pits called Kua (bdnews24.com, 2026)
Natural Resources	Less forest cover than before; streams with less water than elders remember from the past (Datta et al., 2025)
Environmental Problems	Cutting of trees on upper slopes; stone extraction; soil washing away along stream banks (The Daily Post, 2024)
Water Structures	PHED-installed tube wells that stopped working after 1-1.5 years; a water treatment plant that was abandoned (bdnews24.com, 2026)
Health Facilities	One community health worker; Bandarban Sadar Hospital is far but accessible

Important Findings from the Transect Walk:

The transect walk showed that the Suwalok Stream, which used to have water all year round, now becomes very low during dry months (bdnews24.com, 2026). Community members showed us how they dig shallow pits (Kua) right next to the stream where water seeps through sand and soil. These pits have no protection – they are open to dust, animal waste, and water from upstream bathing and washing (The Daily Post, 2024).

One elderly resident, Usaiang Marma (age 50), shared how things have changed: "Back then, there were fewer families, fewer people. Dense forests surrounded us. The streams flowed well. There was water all year round. The crisis was not as severe as it is now" (bdnews24.com, 2026).

Resource Mapping

Resource mapping means community members create a visual picture or map of their community's assets, resources, and problems (Chambers, 2008).

How We Made the Map:
Community members worked together to draw a map on the ground using local materials like sticks, stones, and leaves. Later we transferred it to paper (Minkler & Wallerstein, 2011). The mapping session included 12 participants from different households.

Main Resources We Identified:

Type of Resource	Specific Resources	Access Problems
Natural	Suwalok Stream, seasonal springs, forest areas	Less water available; trees being cut down (Datta et al., 2025)
Water Structures	1 tube well (partly working), 2 ring wells, many Kua pits	Tube wells stop working after 1-2 years; ring wells have too much iron (bdnews24.com, 2026)
Social	Union Parishad Chairman support, Buddhist temple network, traditional Headman	Few NGOs working in the area
Economic	Jhum farming, some daily wage labor	Water shortage limits farming (Datta et al., 2025)

How Water Access Is Unequal:

The mapping showed that not everyone has the same access to water:

Living close to the stream: Families near the stream have easier access but face higher risk of dirty water (The Daily Post, 2024)

Economic status: Poorer families cannot afford to travel to cleaner sources farther away

Gender: Women are mainly responsible for collecting water. They spend 2-4 hours every day during dry season (Datta et al., 2025)

What We Found (Data Analysis and Findings)

Current Ways People Cope with Water Shortage

Based on our participatory analysis, community members use several methods to cope during summer months (bdnews24.com, 2026).

Main Method: Using Kua (Hand-Dug Pits)

The most common coping method is digging shallow pits beside streams where groundwater seeps upward. As Jony Marma, a community resident, told us: "We collect drinking water in the morning... We mostly drink water from the Kua. We don't boil it. Except during the monsoon, it is our only source" (bdnews24.com, 2026).

Other Methods:

Waiting in line at the few working tube wells (some families wait 1-2 hours every day)
Collecting water from the Buddhist temple's ring well (this source is also becoming unreliable)
Collecting rainwater during monsoon (few families have storage systems)
Using less water during the driest months (The Daily Post, 2024)

Water Quality and Health Risks

Even though residents say they rarely get sick from Kua water, doctors warn about serious risks. Dr. Dilip Chowdhury, Resident Medical Officer at Bandarban Sadar Hospital, said: "Drinking water directly from a Kua is not safe under any circumstances. People may have no choice, but they should filter the water when collecting it and boil it before drinking" (bdnews24.com, 2026).

We identified several ways water gets contaminated (The Daily Post, 2024):

Open pits exposed to dust and animal waste

People bathing and washing clothes upstream

Rainwater running into the pits

No filtering or boiling (85% of participants said they do not boil water)

3.3 Barriers to Getting Safe Water

Technical Barriers:
Engineers from the Department of Public Health Engineering admit that normal tube well technology does not work in this area. Executive Engineer Anupam Dey said: "Despite long efforts, the water crisis in Suwalok cannot be resolved. Water cannot be found in many places here. This is an unsuccessful zone" (bdnews24.com, 2026).

Environmental Barriers:
Cutting down trees is a major cause of declining water availability (Datta et al., 2025). Trees and their roots naturally hold water in the soil. When trees are removed, the land cannot hold water as well. Mahbubul Islam, Chief Scientific Officer at the Soil Conservation and Watershed Management Centre, explained: "Trees and stones in the topsoil are crucial. They help retain water naturally. But these sources are being destroyed day by day" (bdnews24.com, 2026).

Political and Economic Barriers:
A water treatment plant project started in 2011 costing Tk2.15 crore was abandoned. A second attempt costing Tk9.46 crore stopped working on April 12, 2024. In total, Tk11.5 crore was spent across two phases without providing any water to the community. This shows serious problems in how projects are managed (bdnews24.com, 2026).

Community Strength and Traditional Knowledge

Despite all these barriers, the Marma community shows remarkable strength (Datta et al., 2025). Traditional knowledge about finding water – reading soil moisture, identifying indicator plants – has been preserved and still guides where people dig pits. The community has also developed social ways of sharing water during shortage, with families taking turns at limited sources (The Daily Post, 2024).

Action Plan (What We Plan to Do Next)

Based on our participatory analysis done together with community members, we developed the following action plan (Minkler & Wallerstein, 2011).

Short-Term Plan (0-6 Months)

Action	Who Will Do It	What Is Needed	How We Measure Success
Install water filters in each home	Community members with NGO help	Ceramic filters (about 3,500 BDT per household)	50% of families using filters
Encourage boiling water	Community health volunteers	Fuel-efficient stoves, metal pots	Fewer diarrhea cases
Bring water by truck during driest months	Union Parishad, PHED	Budget (about 100,000 BDT)	All 160 families get water
Distribute rainwater collection kits	Partner NGOs	100-liter storage containers	80 families have storage

Most Important Plan: Deep Ring Wells

The most important recommendation, strongly supported by community members, is installing deep ring wells that go down 600 feet. As local representative Suikyahla Marma explained: "If new ring wells are installed, they must go 600 feet deep. Otherwise, we will have to install a deep tube well on the other side of the stream. An application has already been submitted" (bdnews24.com, 2026).

Staying Involved After the Project Ends

To make sure the impact lasts, the research team promised to:

Return every 3 months for follow-up visits (Minkler & Wallerstein, 2011)

Stay in touch with the Union Parishad Chairman

Share this report with relevant government departments

Connect the community with NGOs working on water and sanitation

5. Conclusion

This participatory research study shows that the Marma community of Amtoli Para faces a severe and worsening water crisis during summer. They are forced to rely on unsafe Kua pits for drinking water (bdnews24.com, 2026). Although the community has developed impressive coping methods based on traditional knowledge, structural barriers – including wrong technology choices, cutting down trees, and failed government projects – prevent long-term solutions (Datta et al., 2025; The Daily Post, 2024). The voices of community members, especially women who carry the main burden of water collection, urgently call for deeper ring wells, accountable water infrastructure management, and recognition of their traditional water knowledge (bdnews24.com, 2026).

This research confirms a core principle of participatory research: communities are not passive victims but active agents with valuable knowledge and clear solutions (Cornwall & Jewkes, 1995; Freire, 1970). What the Marma community needs is not outside experts imposing solutions, but accountable partners who listen to their insights and act on their priorities (Chambers, 2008; Minkler & Wallerstein, 2011).

Pictures : In search of pure water

References:

bdnews24.com. (2026, March 31). Thirst grips Bandarban hills as unsafe water pits replace reliable sources. https://bdnews24.com/bangladesh/64c0374bb983

Chambers, R. (2008). Revolutions in development inquiry. Earthscan.

Cornwall, A., & Jewkes, R. (1995). What is participatory research? Social Science & Medicine, 41(12), 1667–1676.

Datta, R., Hurlbert, M., Kibria, A., Singha, R., Kairy, B., & Chattapadhya, S. (2025). Flood disaster impact and adaptation: Learning reflections from Marma Indigenous community in Bandarban. In Climate change and Indigenous communities (pp. xx–xx). Taylor & Francis Group.

Food and Agriculture Organization of the United Nations. (2017). Discovery learning on land and water management: A practical guide for farmer field schools. TAPipedia. https://tapipedia.org

Freire, P. (1970). Pedagogy of the oppressed. Continuum.

Israel, B. A., Schulz, A. J., Parker, E. A., & Becker, A. B. (2017). Community-based participatory research. In R. J. DiClemente, L. F. Salazar, & R. A. Crosby (Eds.), Health behavior theory for public health (2nd ed., pp. 245–272). Jones & Bartlett Learning.

Isubikalu, P. (2007). Stepping-stones to improve upon functioning of participatory agricultural extension programmes: Farmer field schools in Uganda (Publication No. 12345) [Doctoral dissertation, Wageningen University]. AGRIS/FAO.

Minkler, M., & Wallerstein, N. (Eds.). (2011). Community-based participatory research for health: From process to outcomes (2nd ed.). Jossey-Bass.

The Daily Post. (2024, May 3). Fresh water crisis in Bandarban. https://www.dailypost.net/national/6mh